

JECRC UNIVERSITY JAIPUR
Session: 2025-26
Subject: Mathematics-II (DMA022A)
Course: B. Tech - II Semester
Tutorial Sheet – 12
UNIT-2

M.M:64

PART – A

(5x2=10)

- A1 Find the integrating factor of $\frac{dy}{dx} + \frac{3}{x}y = x^2$.
- A2 The equation $\frac{dy}{dx} = \frac{x+y}{x}$ can be solved by separaton of variables / exact method/ Substitution ($y = vx$)/ Bernoulli.
- A3 Check whether the equation is exact: $(2xy + 3)dx + (x^2 + 4y)dy = 0$.
- A4 The order and degree of the equation $\left(\frac{dy}{dx}\right)^3 + \left(\frac{dy}{dx}\right)^2 = x$ is.....
- A5 Solve the differential equation $\left(\frac{dy}{dx}\right)^2 - 4\frac{dy}{dx} + 3 = 0$.

PART – B

(3x7=21)

- B1 Solve $3x(1-x^2)y^2 \frac{dy}{dx} + (2x^2-1)y^3 = ax^3$.
- B2 Solve $(1+y^2) + (x - e^{-\tan^{-1}y}) \frac{dy}{dx} = 0$.
- B3 Solve $4yp^2 + 2px - y = 0$.

PART – C

(3x11=33)

- C1 Solve $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$.
- C2 Solve $(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)xdy = 0$.
- C3 Solve (i) $\sin px \cos y = \cos px \sin y + p$ (ii) $x - yp = ap^2$.

Answers:

A1	x^3	A2	By substitution	A3	exact	A4	O=1, D=3.	A5	$(y-x-c)(y-3x-c) = 0$
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B1 $y^3 = ax + cx\sqrt{1-x^2}$	B2 $xe^{-\tan^{-1}y} = \tan^{-1}y + c$	B3 $Y^2 = ax + a^2$
C1 $x^3y^2 + x^2 / y = c$	C2 $xy + \log \frac{x}{y} - \frac{1}{xy} = c$	C3 (i) $y = cx - \sin^{-1}c$ (ii) $x = \frac{p}{\sqrt{1-p^2}}(c + a \sin^{-1} p),$ $y = -ap + \frac{1}{\sqrt{1-p^2}}(c + a \sin^{-1} p)$